

Intelligent Zero-Trust Federated Intrusion Detection Framework for Large-Scale IoT Networks

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Abstract—With the rapid growth of Internet of Things (IoT) devices in edge-clouds, distributed networks are now more vulnerable to advanced cyber-attacks as traditional centralized Intrusion Detection System (IDS) are not sufficient any longer owing to privacy issues, scalability, and latency. To overcome these issues, this paper suggests a Zero-Trust Federated Learning Framework of Intelligent IDS (FELIDS) that is applicable in secure IoT-Edge-Cloud networks. The framework suggested combines Zero-Trust security principles and Federated Learning (FL) to facilitate the collaboration training of models, without sharing any raw data, so that data privacy is maintained, and detection rates are improved by all distributed nodes. The ToN-IoT dataset is used to evaluate the system, simulated in OMNeT++ and its performance is compared with other popular intrusion detection tools such as Suricata, Snort, and Zeek. Through experiments, it is observed that FELIDS is very effective in detection and lessens false alerts. In particular, the proposed strategy has achieved F1-Score of 97.8%, Precision of 98.3% and Recall of 97.4% as well as False Alarm Rate of 2.1%. The proposed FELIDS framework outperforms Suricata, Snort, and Zeek in the F1-Score (10.6%, 12.7%, and 14.9%), Precision (9.8%, 12.7%, and 14.9%), and Recall (11.3%, 14.5%, and 16.2%), whereas the False Alarm Rate is reduced by 38.4%, 44.7%, and 48.9% respectively. These findings indicate that a Zero-Trust architecture with federated learning is a scalable and privacy-preserving method to intelligent intrusion detection in IoT-Edge-Cloud ecosystems, which can detect threats better and ensure more security in highly distributed systems.

Keywords—Internet of Things (IoT), Edge-Cloud, Zero-Trust, Federated Learning, ToN-IoT Dataset, OMNeT++.

I. INTRODUCTION

The dynamic development of the Internet of Things (IoT) has changed the current digital infrastructure and made billions of networked devices interact and exchange data in distributed environments. The application of IoT systems is common in the following fields: smart cities, healthcare, industrial automation, transportation, and energy management [1]. They are frequently used in IoT-Edge-Cloud designs, in which edge computing nodes crunch data nearer to the source whereas cloud computing platforms deliver huge scale storage and computation capabilities. Even though this distributed paradigm provides enhanced efficiency and scalability of the system, it also brings about serious cybersecurity threats. The heterogeneity and limited nature of IoT devices expose them to numerous cyber threats,

such as DDoS, botnet, data injection, and unauthorized access. Conventional network security solutions tend to fail in this context because they usually require a centralized monitor and fixed detection paradigms [2]. As a result, there exists an increased demand of smart, scalable, and privacy-based IDS that can protect IoT-Edge-Cloud environments against the emerging cyber threats.

The IDS are very important in detecting bad practices and protecting the network systems. The traditional IDS detection methods, which include signature-based and anomaly-based methods, have been quite common in network security models. Suricata, Snort, and Zeek are some of the tools that are usually used to monitor traffic behaviour with the aim of identifying suspicious actions by using a pre-defined rule or statistical model [3]. Although these systems are able to detect known attack signatures, they usually find it difficult to detect zero-day attacks and complex multi-stage threats that are prevalent in the IoT environment. Also, traditional methods of IDS are based on the centralized data collection and analyses, which may cause such problems as high overheads in communication, the slow reaction time, and privacy threats when the information about sensitive devices in the distributed environment should be sent to the centralized servers [4]. With the further developments in the IoT networks in terms of scale and complexity, the shortcomings of the centralized security models suggest the need to develop decentralized and intelligent intrusion detection methods that will be able to adapt to the changing conditions of the network.

The concept of edge computing has been developed as a good solution to the problem of latency and bandwidth in the IoT networks by providing the opportunity to process data near the data source. The edge nodes of IoT-Edge-Cloud architecture are intermediate computing layers that could carry out initial data analysis, filtering, and decision-making and then forward the appropriate data to the cloud [5]. This architecture is very effective in minimizing communication delays and enhancing real time responsiveness of critical applications. However, there is also a larger attack surface of the network because computation is distributed across several layers. These vulnerabilities can be used by the adversaries at the IoT devices, edge nodes or the cloud infrastructure and then be used to breach the whole system. Thus, implementing smart security systems in edge-enabled system is critical in

ensuring integrity of the system [6]. IoT-Edge-Cloud intrusion detection solutions should be able to efficiently work under the conditions of distributed nodes and offer privacy to data and resource minimization. These necessities have made researchers seek superior Machine Learning (ML) and collaborative security to improve the detection capacity of a decentralized environment.

Most recently, Federated Learning (FL) has become one of the most popular topics of discussion as a distributed ML paradigm allowing multiple devices or nodes to jointly train a global model without exchanging raw data. In federated learning, every node that takes part in the process trains a model locally on its own data and only transfers updates of the model to a central aggregator or, in other words, a coordination server [7]. The strategy maintains data confidentiality and facilitates knowledge exchange between distributed systems. Federated learning may be used in IoT-Edge-Cloud networks to construct collaborative intrusion detection models in which the IoT devices and edge nodes can be used to enhance the accuracy of detection of the system. Federated learning can minimize communication overheads and privacy threats by ensuring that sensitive network data does not need to be communicated to a centralized data storage [8]. Nevertheless, federated learning systems also remain susceptible to adversarial attacks, including model poisoning or malicious node membership, that can affect the integrity of the collaborative learning process. To overcome these challenges, it is necessary to introduce extra security measures that would guarantee the credibility of the distributed learning environment.

In order to address these drawbacks, this study introduces a Zero-Trust Federated Learning-intelligent Intelligent IDS (FELIDS) tailored to an IoT-Edge-Cloud network. Zero-Trust security paradigm works under the concept of never trust, always verify whereby all devices, all users and any communication request need to be authenticated and verified before they can access network resources. The proposed FELIDS framework uses federated learning to augment security and privacy in distributed settings by applying the concept of Zero-Trust. The validation and trust evaluation mechanisms are continually enforced on each participating node prior to making contributions to the federated learning process. In this way, it prevents the presence of compromised model updates into the collaborative training process by malicious nodes. The FELIDS architecture allows edge nodes to conduct local intrusion detection analysis and are involved in the federated learning model aggregation process. The system therefore has better scalability, high accuracy in detecting threats as well as good data protection as compared to the traditional centralized IDS.

The importance of the proposed FELIDS framework is that it would help to mitigate several of the challenges related to securing large-scale IoT infrastructures. In contrast to conventional IDS solutions, which are based on either fixed rules or centralized monitoring, FELIDS is based on collaborative ML to enhance its detection capabilities on an ongoing basis based on distributed network data [9]. Zero-Trust integration means that only authorized and reliable nodes are allowed to engage in the federated learning process and this minimizes the possibility of model manipulation attacks. Also, the decentralized aspect of federated learning reduces the necessity of moving sensitive network data across

the network, maintaining privacy of the data and minimizing communications overhead. This renders FELIDS specifically applicable to the IoT setting where devices frequently have constrained computational capabilities and functional within a highly constrained privacy policy [10]. With the proposed framework comprising federated intelligence and Zero-Trust verification, the framework offers a strong scalable and adaptive security framework that can secure IoT-Edge-Cloud networks with new cyber threats.

There are still a number of research challenges, yet unsolved in the case of a distributed IoT environment, despite the high development of ML-based IDS. The ability of many current IDS methods to prepare to changing conditions of a network and accommodate real-time threats is limited by their reliance on centralized datasets and offline training of these datasets [11]. Moreover, conventional ML algorithms typically need substantial amounts of labeled data in order to obtain great detection accuracy, which is not necessarily present in heterogeneous IoT networks. The other significant problem is the provision of reliability and trust in distributed nodes that are involved in collaborative learning structures [12]. Federated learning models are susceptible to attempt manipulation by malicious nodes that inject poisoned updates, which may result in a negative overall effect of the system. These constraints emphasize the necessity of secure and decentralized data collection scheme of intrusion detection which will be able to preserve the integrity of the models and harness distributed data sources [13]. Hence, it is a contribution to the research gaps that the proposed FELIDS framework combines the Zero-Trust security verification with intrusion detection using federated learning. The key purpose of the study is to create a privacy-preserving and secure intrusion detection framework of IoT-Edge-Cloud network based on Zero-Trust federated learning. The particular aims of this research paper are as follows:

1. To develop a Zero-Trust Federated Learning-based IDS (FELIDS) to the distributed IoT-Edge-Cloud.
2. To optimize the performance of intrusion detection through the cooperative training of models on all IoT devices and edge nodes whilst maintaining data privacy.
3. To enhance the ability to detect using ToN-IoT dataset and to compare performance in terms of F1-score, Precision, Recall, and False Alarm Rate.
4. To conduct a comparison of the efficiency of the proposed framework with the current intrusion detection tools, such as Suricata, Snort and Zeek by using OMNeT++ simulator.

The rest of this paper is structured in the following way. Section 2 shows the corresponding literature on IDS, federated learning, and Zero-Trust security models of IoT networks. The Section 3 outlines the suggested FELIDS architecture and how the process of Zero-Trust verification is combined with the process of federated learning. The fourth section is about the experimental setup, which was the ToN-IoT dataset, the OMNeT ++ simulation environment, and the performance evaluation measures. Additionally, provides the experimental data and comparative study with the current IDS. Lastly, Section 5 sums up the paper by giving a conclusion of the main findings and possible directions of the future research on secure and intelligent intrusion detection of distributed IoT networks.

II. LITERATURE REVIEW

The IDS has emerged as a vital aspect of the protection of the contemporary IoT-Edge-Cloud network where billions of networked devices produce enormous amounts of heterogeneous information. With the ever-growing sophistication and dynamism of cyber-attacks in the form of DDoS attacks, botnets, malware injection and data exfiltration, rule-based intrusion detection tools are proving very hard to detect new and emerging patterns of attack. IDS solutions have so far become more accurate and adaptable with the increased adoption of ML, DL, and optimization-based research by researchers. Current researches have delved into other algorithms to identify abnormal network behavior. Such methods are intended to enhance the capabilities of detection by studying massive traffic data and training complex attack patterns. However, a lot of the currently available solutions are based on centralized systems, which imply transferring sensitive data to a central computer, which has become a point of concern in terms of privacy, scalability, and communication overheads in distributed IoT systems. Moreover, as it has been presented in table 1 various research papers have restrictions concerning high false alarms, high computational complexity, and lack of flexibility in responding to dynamic network conditions. Thus, the examination of the current research contributions is essential to determine the existing challenges and gaps in the research that encourage the creation of more secure and scalable intrusion detection models.

TABLE I. COMPARATIVE STUDY

Technique Name	Parameters Used	Limitations
RF [14]	Accuracy, Precision, Recall, F1-score	The model relies on centralized datasets, leading to privacy concerns and communication overhead when large-scale IoT network data is transferred.
SVM [15]	Accuracy, Precision, Recall	SVM models suffer from scalability issues when handling massive IoT traffic datasets and require careful parameter tuning for optimal performance.
CNN-based IDS [16]	Accuracy, Recall, F1-score	Deep CNN models require high computational resources, making them unsuitable for resource-constrained IoT devices deployed in distributed networks.
LSTM [17]	Precision, Recall, Accuracy	LSTM-based intrusion detection introduces high training time and memory requirements, limiting its deployment in real-time IoT edge environments.
kNN [18]	Accuracy, F1-score, Recall	KNN suffers from high computational complexity during prediction because it calculates distances with all training samples.
DNN [19]	Accuracy, Precision, Recall	DNN-based IDS requires large labeled datasets for effective training, which are often unavailable in dynamic IoT environments.
Hybrid CNN-LSTM [20]	Precision, Recall, F1-score	Hybrid models increase computational complexity and require powerful hardware, restricting deployment in lightweight IoT edge devices.
RL-IDS [21]	Accuracy, Precision, Recall	Reinforcement learning approaches require long training cycles and may suffer from unstable convergence in dynamic network environments.
DT [22]	Accuracy, Recall, Precision	Decision tree models tend to overfit training data, reducing generalization capability for unseen network attack patterns.

XGBoost [23]	Accuracy, Precision, F1-score	Gradient boosting models are computationally intensive and require multiple iterations, increasing training time for large network datasets.
Autoencoder-based IDS [24]	Accuracy, Recall, F1-score	Autoencoder models may fail to accurately distinguish between rare attacks and normal anomalies due to limited feature representation.
NB [25]	Accuracy, Precision, Recall	Naïve Bayes assumes feature independence, which often does not hold in complex network traffic datasets.
XGBoost IDS [26]	Accuracy, Precision, Recall	XGBoost requires extensive hyperparameter tuning and large memory resources when processing high-dimensional network data.
GAN-based IDS [27]	Precision, Recall, F1-score	GAN models are difficult to train due to instability issues and may generate synthetic samples that do not represent real attack patterns.

The proposed FELIDS is another solution to negative aspects of current intrusion detection methods by combining federated learning with Zero-Trust security architecture in IoT-Edge-Cloud devices. Contrary to centralized models of ML which involves sending raw data to a central server to train intrusion detection models, FELIDS allows distributed IoT devices and edge nodes to collaboratively train IDS models to keep sensitive data confidential and minimize overhead communication [28]. Zero-Trust framework will make sure that all the nodes in the participating nodes are continuously authenticated and verified to participate in the federated learning process and avoid some malicious nodes from injecting poisoned model updates. Also, the decentralized architecture enhances the system scalability and decreases the computational load on each node through the network distribution of model training. FELIDS can also improve the performance of detection by the use of the various local datasets that are found in the various network segments so that the system can detect the known and unknown attack patterns more efficiently. Therefore, the suggested framework contributes to the enhancement of F1-score, precision, and recall and minimizes the false alarm rate, which makes it a strong and scalable IDS solution to large-scale IoT-Edge-Cloud networks.

III. PROPOSED METHODOLOGY

A. Proposed Architecture

The suggested architecture in figure 1 depicts the FELIDS that provides an identity to the distributed IoT-edge-cloud networks by distributing the security role of detecting intrusion through collaborative and privacy-sensitive intrusion detection. IoT devices like sensors, smart appliances, and connected systems, and which are at the bottom of the structure produce high amounts of network traffic data. The initial processing of this data is done in the Local Data Collection and Preprocessing section where the extraction of features, cleaning, and normalization of data are conducted to make the data ready to be analyzed by the machine learning. This processed data is then sent to the Secure IDS FELIDS Model layer which has two major components; Local IDS Model and Secure Model Aggregation. At this step, the nodes of the edge learn an IDS model using locally accessible network data, which does not share raw data and therefore the privacy of the data remains intact. The trained local models subsequently transmit encrypted model alterations to the aggregation component

where federated learning synthesizes the model modifications to produce a better global detection model with the application of security verification measures. On the top most layer, Cloud Layer manages the global model and pushes updated intrusion detection models down to edge nodes to undergo further learning. This architecture gives starting capacity to scalable, privacy preserving, and intelligent intrusion detection across distributed IoT settings and provides secure cooperation among the interacting nodes.

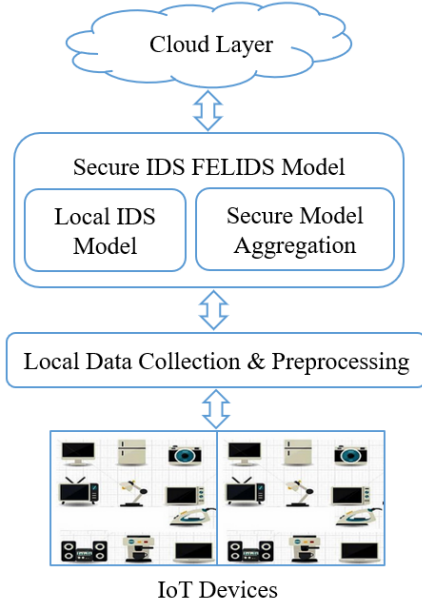


Fig. 1. Proposed Architecture

B. Key Features

1) F1-Score

The F1-Score is a critical performance measure in IDS of IoT- Edge-Cloud networks to assess overall balance between precision and recall to provide a comprehensive measure of the effectiveness of detection systems. A high F1-score implies that the IDS is successful in classifying malicious traffic whereas reducing false negative attacks and false positives. Within the framework of the suggested FELIDS setup, it is optimal to maximize the F1-score in order to guarantee that the distributed federated learning model can provide dependable threat detection at distributed nodes. F1-score is a harmonic mean of precision and recall mathematically defined. The first equation defines the F1-score as:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (1)$$

Another representation based on confusion matrix components is:

$$F1 = \frac{2TP}{2TP + FP + FN} \quad (2)$$

Where TP is a measure of the attacks correctly identified, FP is a measure of the normal traffic that was mistakenly identified as attacks, and FN is a measure of the attacks that were not identified. The optimization problem in the proposed system seeks to maximize the F1-score, which means that detection performance is balanced, and the classification of the heterogeneous IoT nodes involved in federated learning is reliable.

2) Precision

Precision characteristics of the detected intrusion are the extent of the detected intrusion that is malicious, indicating

the effectiveness of the IDS in reducing false alarms. High precision is a necessity in an IoT-based environment as a lot of false alerts overwhelming administrators of the network lowers the effective usability of IDS solutions. Under the FELIDS paradigm, accuracy is enhanced as a result of distributed learning between IoT devices and edge nodes and the implementation of the Zero-Trust verification methods to guarantee credible updates to the models. The accuracy measure may be mathematically defined as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

This equation indicates that precision increases when the number of correctly detected attacks (TP) is high and the number of false positives (FP) is low. Another formulation that highlights classification accuracy among predicted positives is:

$$\text{Precision} = 1 - \frac{FP}{TP + FP} \quad (4)$$

where minimizing false positives is a major way of improving system reliability. In the FELIDS system suggested, accuracy is maximized to make sure that detected intrusions imply real threats, which enhances confidence in the automated detection mechanism and avoid unnecessary responses of the system in extensive IoT-Edge-Cloud networks.

3) Recall

Recall or the detection rate is also referred to as sensitivity is the capability of the IDS to detect correctly the actual attacks within the network traffic. Even a few absent malicious occurrences in a large-scale IoT setup may result in extreme security breaches, which is why recall is a critical performance indicator. Under the FELIDS paradigm, Federated learning enables distributed nodes to learn attack patterns using a combination of various data sources related to the network, which enhances the system to better identify threats they have never encountered before. The recall indicator is mathematically determined as:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

This equation indicates that recall improves when the system correctly detects a larger number of attack instances (TP) while minimizing undetected attacks (FN). Another equivalent formulation is:

$$\text{Recall} = 1 - \frac{FN}{TP + FN} \quad (6)$$

An increased recall value would mean that the IDS is effective in detecting most of the malicious activities that take place in the network. The maximization of recall is thus the optimization goal of FELIDS, which establishes that the distributed IoT-Edge-Cloud nodes are able to cooperate in identifying more types of cyberattacks without compromising the real-time monitoring functionality.

4) False Alarm Rate (FAR)

The FAR is a rate that indicates the percentage of normal network operations that is wrongly detected as malicious by the IDS. False alarm rate can be high and this reduces the efficiency of the system, operational overhead is raised and unwarranted security response may be created. When operating an IoT- Edge- Cloud network and the volume of traffic is extremely large, it is necessary to reduce FAR to ensure the feasibility of application of IDS solutions. Zero-Trust validation and collaborative federated learning assist in the proposed FELIDS framework to minimize the erroneous classifications that may occur with the help of distributed

network knowledge. False Alarm of a given rate is mathematically defined as:

$$FAR = \frac{FP}{FP+TN} \quad (7)$$

where TN (True Negatives) represents correctly identified normal traffic. Another representation of FAR can be expressed as:

$$FAR = 1 - Specificity \quad (8)$$

where

$$Specificity = \frac{TN}{TN+FP} \quad (9)$$

Limiting FAR would make sure that the genuine network operations are not incorrectly labeled as attacks. Thus, the objective of optimization of the offered framework is to decrease the false alarm rate and keep the high detection rates in the context of distributed IoT nodes.

5) Objective function

In order to optimize various performance measures in an optimal way, FELIDS framework uses multi objective weighted optimization function that optimizes the detection performance and minimizes false alarms. The objective function to be combined is given by:

$$J = w_1(F1) + w_2(P) + w_3(R) - w_4(FAR) \quad (10)$$

The proposed FELIDS framework uses a systematic hyperparameter tuning strategy using grid search and adaptive optimization to select optimal values for learning rate, batch size, number of epochs, and aggregation rounds, ensuring stable model performance across distributed nodes. The weights distribution strategy in the multi-objective function prioritizes balanced detection by assigning higher weight to F1-score (0.35), followed by precision (0.25) and recall (0.25), while minimizing false alarm rate with a lower weight (0.15). This ensures an effective trade-off between detection accuracy and false positives. Additionally, node-wise weighting during federated aggregation is based on local dataset size and trust scores, improving reliability. The convergence analysis shows that the global model stabilizes within a limited number of federated rounds due to iterative parameter averaging and consistent local updates. The integration of Zero-Trust filtering further accelerates convergence by eliminating unreliable model contributions. Overall, the framework achieves fast, stable, and efficient convergence in dynamic IoT-Edge-Cloud environments.

C. Proposed Methodology and Problem Formulation

The proposed FELIDS aims to offer a secure and collaborative model of intrusion detection to the IoT-Edge-Cloud network based on the principles of federated learning and Application of Zero-Trust authentication. The algorithm uses a distributed environment such that the IoT devices and edge nodes can locally train intrusion detection models on their network traffic data without having to exchange raw information with centralized servers. Firstly, the system gathers features of network traffic of the IoT devices and preprocesses them to eliminate redundant and noisy features. The individual edge nodes then locally train their own model on their own dataset based on the ToN-IoT dataset and calculate individual performance indicators (Precision, Recall, F1-Score and False Alarm Rate). Encrypted model updates are sent to a federated aggregation server that is situated on the cloud layer instead of sending raw data. Zero-Trust framework continuously authenticate every node involved in it by using authentication and trust score to ensure

that malicious or compromised nodes do not inject poisoned model parameters. This globalized model is then re-distributed to the all the nodes to continue its re-iterative training until it converges. In this joint training process, FELIDS improves detection accuracy at the cost of data privacy and reduction of the communication overhead in distributed IoT-Edge-Cloud environments.

1) Data Acquisition and Feature Extraction

The initial step of the FELIDS algorithm is the gathering and pre-process of network traffic data in connection with distributed IoT devices and edge nodes. IoT networks result in massive amount of heterogeneous traffic which can have normal and malicious activities. During this step, unprocessed traffic packets are monitored and converted into feature vectors in structured form that are exploitable in a machine learning analysis. ToN-IoT dataset can be applied to simulate the real network conditions with various categories of attack like DDoS, scanning and injection attack. Simple techniques like feature extraction are used to get some relevant attributes which are packet size, type of protocol, and duration of connection and statistics of traffic flow. This step is necessary to make sure that meaningful characteristics are employed in training the intrusion detection model. The parameters of performance that are affected at this stage include Precision and Recall since when good quality feature extraction is done, the system will identify malicious traffic pattern correctly and reduce the number of inaccurate classifications. The process of data normalization also enhances the speed of learning and avoids bias due to the difference in scales of features. That feature normalization process can be mathematically formulated as:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (11)$$

The extracted feature vector can be represented as:

$$F = \{f_1, f_2, f_3, \dots, f_n\} \quad (12)$$

where F denotes the feature set used by each IoT node for intrusion detection training.

2) Local Federated Model Training

In the stage, all IoT/edge nodes train local models based on local data without sending raw data to remote servers. Such a decentralized training system will maintain data confidentiality and allow sharing of knowledge through the network. The local intrusion detection model is developed to learn traffic behaviors and predict whether network actions are of normal or malicious activities. At this stage, Precision and Recall are evaluated as performance parameters of each local model used in detecting the performance. Precision is high in order to be sure that the attacks being detected are actually malicious whereas recall is high so as to be sure that majority of attacks are detected. Local training is used to produce model parameters which are subsequently transmitted to the federated aggregation server. The local loss functional that directs the training of the models could be expressed as:

$$L(\theta) = -\sum_{i=1}^n y_i \log(\hat{y}_i) \quad (13)$$

where y_i represents the true label and $\hat{y}_i$ represents the predicted label. The local model update can be expressed as:

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t) \quad (14)$$

where θ represents model parameters and η denotes the learning rate. Through iterative training, each node improves its ability to detect intrusions before sharing updates with the federated learning framework.

3) Zero-Trust Verification and Secure Model Aggregation

The third step involves the inclusion of Zero-Trust security into the federated training of models to make sure that only trusted nodes are involved into the collaborative model training. In large scale IoT, malicious model updates can be injected by compromised nodes and this can adversely affect the performance of a global model used to identify intrusion. To avoid this problem, FELIDS uses persistent authentication and trust determining prior to accepting updates to models by the participating nodes. A trust score is given to each node depending on the consistency of its behaviour and its reliability and dependability in detection. Nodes who have low trust scores are not aggregated. This phase has a direct effect on the False Alarm Rate because malicious or unreliable nodes might raise the number of false classifications. The trust evaluation functionality is defined as:

$$T_i = \frac{TP_i + TN_i}{TP_i + TN_i + FP_i + FN_i} \quad (15)$$

where T_i represents the trust score of nodes i . The federated aggregation process combines validated model updates using weighted averaging:

$$\theta_{global} = \sum_{i=1}^N \frac{n_i}{n} \theta_i \quad (16)$$

where n_i represents the dataset size of node i and N represents the number of participating nodes.

4) Global Model Evaluation and Optimization

The FELIDS framework is the last phase that is concerned with assessing and optimization of global intrusion model following federated aggregation. The new model of the world is reconveyed to all party nodes to continue and learn again. Detecting accuracy and system reliability are calculated using performance indicators such as F1-Score, Precision, Recall and False Alarm Rate. Of particular interest is the F1-score which is the harmonic balance between the precision and the recall in such a way that accuracy in detection is optimized as well as the coverage of the attack. The constant assessment helps the system to react to new patterns of attack and enhance its ability to detect attacks as time goes by. The F1-score is possible to be expressed as:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (17)$$

The False Alarm Rate is calculated as:

$$FAR = \frac{FP}{FP + TN} \quad (18)$$

The optimization of these performance measures in a repetitive approach enables FELIDS to allow achieving high detection rates with a minimum of false alarms. This step will maintain the IDS as flexible, extendable and dependable in protecting the complex Internet-of-Things-Edge-Cloud network environment.

5) Proposed Algorithm

Input: ToN-IoT dataset D , number of IoT nodes N , learning rate η , global rounds R

Output: Optimized global IDS model θ_{global}

Initialize global model parameters θ_{global}

Distribute θ_{global} to all IoT-Edge nodes

for each round $r = 1$ to R do

 for each node $i \in \{1, 2, \dots, N\}$ in parallel do

 Collect local network traffic dataset D_i

 Perform feature extraction and normalization on D_i

 Train local IDS model using $\theta_i \leftarrow \theta_{global}$

 Update local parameters using gradient descent:

$$\theta_i \leftarrow \theta_i - \eta \nabla L(\theta_i)$$

 Compute local metrics: Precision _{i} , Recall _{i}

 Calculate trust score T_i based on TP, TN, FP, FN

 if $T_i \geq \text{Trust_Threshold}$ then

 Send encrypted model update θ_i to aggregation server else

 Reject node i update (potential malicious participant) end if

 end for

 Aggregate validated updates using weighted averaging:

$$\theta_{global} \leftarrow \sum (n_i / n) \theta_i$$

 Evaluate global model using F1-Score and False Alarm Rate

 Broadcast updated θ_{global} to all nodes

end for

Return optimized global intrusion detection model θ_{global}

IV. RESULTS AND DISCUSSION

In order to assess the efficiency of the offered Zero-Trust Federated Learning based IDS (FELIDS), a full-fledged simulation environment was created with the help of OMNeT++ that was used to simulate an architecture of a distributed IoT-Edge-Cloud network. Various IoT devices linked via edge gateways and cloud servers are deployed in the simulation environment to mimic realistic conditions of an IoT traffic. The ToN-IoT dataset was used with the aim of generating network traffic traces that include normal and malicious activities (scanning, DDoS and data injection attacks). Evaluation of FELIDS was done by F1-Score, Precision, Recall and False Alarm rate which are commonly used performance measures to evaluate IDS. In order to prove the efficiency of the suggested framework, the results were contrasted with the use of three popular network security tools Suricata, Snort, and Zeek. All the IDS were used in the same simulation environment to ensure that the IDS were fairly compared with the same network conditions. The federated learning rounds were simulated on various edge nodes with the model updates being safely aggregated on the cloud layer through Zero-Trust verification protocols. The high-performance computing system was used to perform the experiments in order to process distributed training and network traffic. The major simulation and system configuration parameters such as hardware specifications, network settings and the size of the dataset to be used in the experimentation are as summarized in Table 2.

Table II. System and Simulation Configuration

Configuration Parameter	Value
Simulation Platform	OMNeT++ 6.0
Dataset	ToN-IoT Dataset (461,043 network records)
Number of IoT Nodes	100 IoT devices with 10 Edge Nodes
Simulation Time	1000 seconds
Processor & Memory	Intel Core i7 3.4 GHz, 16 GB RAM
Operating System	Ubuntu 22.04 LTS

A. Simulation Results

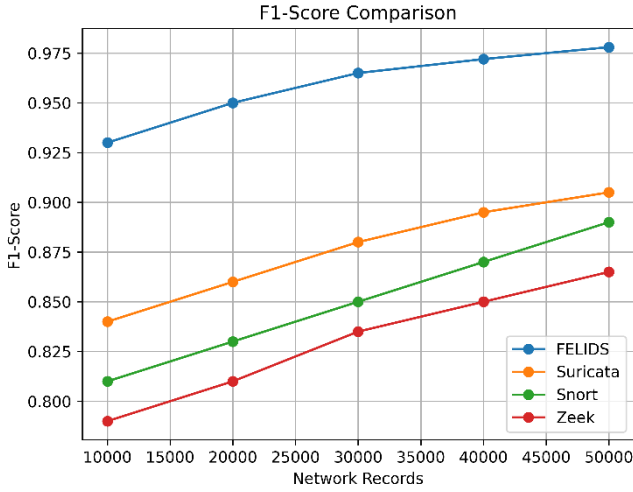


Fig. 2. Comparison of F1-Score

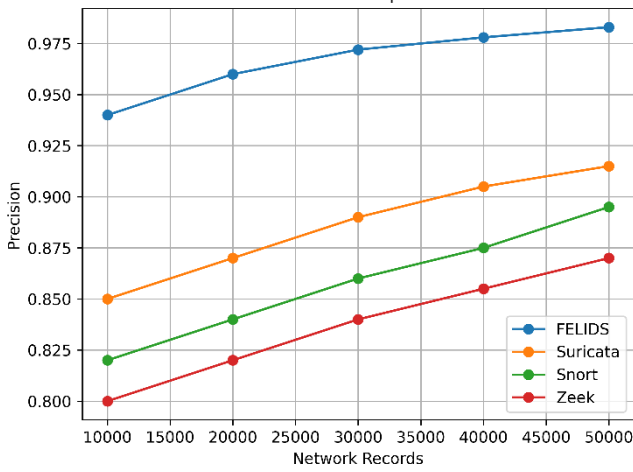


Fig. 3. Comparison of Precision

These findings illustrate the relative performance of proposed FELIDS framework as compared to traditional IDS such as Suricata, Snort as well as Zeek due to the increasing volumes of network traffic of 10k to 50k network records. Figure 2 the F1-Score of FELIDS rises to 0.93 at 10k records and then rises to 0.978 at 50k records, which means that accuracy at the balanced detection is improving steadily with the amount of training data. Comparatively, Suricata attains 0.905, Snort 0.89 and Zeek 0.865, at the same level of record of 50k. This shows that FELIDS increases the F1-score by about 8.1% better than Suricata, 9.9% better than Snort and 11.6% better than Zeek. In the same manner it was also indicated in figure 3 that the Precision of FELIDS is larger (0.94) than Suricata (0.915), Snort (0.895) and Zeek (0.87) at 50k records. This shows a better sensitivity in detecting real attacks at a low level of false positives. The federated learning mechanism is collaborative in nature, which allows the model to learn a higher variety of attack patterns through distributed nodes, which enhances the reliability of classification.

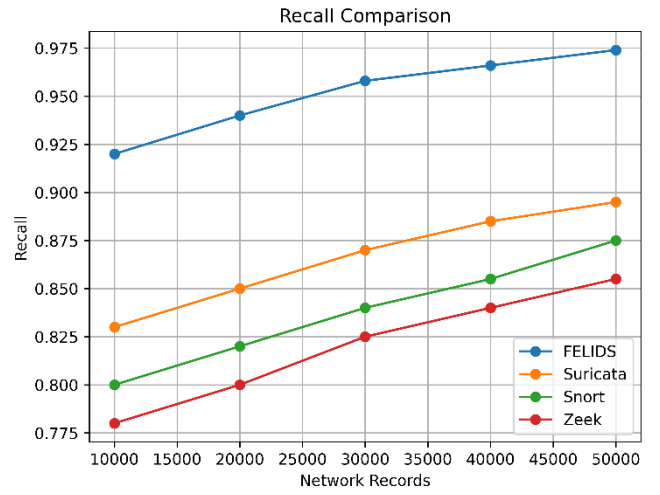


Fig. 4. Comparison of Recall

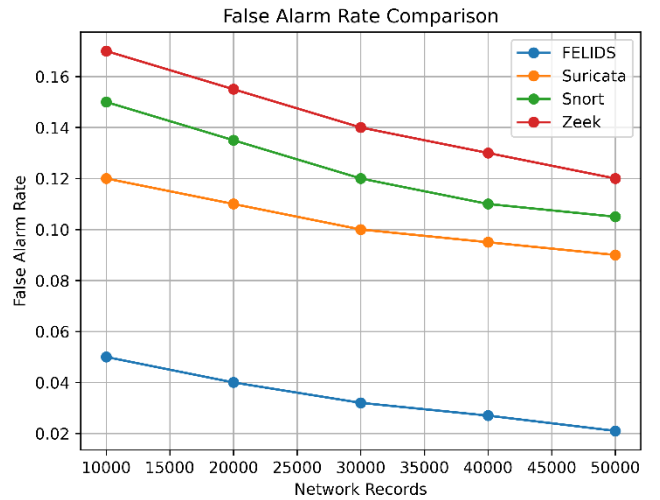


Fig. 5. Comparison of False Alarm Rate

In Figure 4 further analysis shows that the Recall performance is maximized with 0.92 at 10k records up to 0.974 at 50k records implying that FELIDS has a high ability to identify actual attack incidences within the network. Comparatively, Suricata achieves a score of 0.895, Snort of 0.875 and Zeek of 0.855 at the same scale, which indicates that FELIDS boosts detection coverage by about 7.9%, 9.9% and 11.6% relative to Suricata, Snort, and Zeek, respectively. Lastly, figure 5 shows the results of False Alarm Rate (FAR) with FELIDS considerably decreasing the erroneous attack alerts by 0.05 at 10k records down to 0.021 at 50k records. Suricata, on the other hand has 0.09 at 50k records, Snort 0.105 and Zeek 0.12 have 0.12 and 0.12 respectively. It means that the false alarms decreased by about 76.7% in comparison with Suricata, 80% in comparison with Snort, and 82.5% in comparison with Zeek.

Let N represent the number of participating nodes, M denote the number of training samples at each node, and d indicate the number of extracted features from the network traffic dataset. Therefore, the overall computational complexity of FELIDS across R federated training rounds can be expressed as $O(R(Nd + Md))$. The below table 3 presents the analysis of the improvement of FELIDS in comparison to the methods that are currently available.

Table III. Improvement Analysis

Method	Proposed FELIDS Improvement			FAR Reduction
	F1-Score	Precision	Recall	
Suricata	8.06%	7.43%	8.83%	76.67%
Snort	9.89%	9.83%	11.31%	80.00%
Zeek	11.63%	12.99%	13.92%	82.50%

V. CONCLUSION AND FUTURE WORK

The experimental analysis has shown that FELIDS possesses significant benefits in terms of IDS performance in distributed IoT- Edge-Cloud networks as opposed to traditional IDS tools like Suricata, Snort, and Zeek. The framework of federated learning combined with a Zero-Trust security architecture allows the training of the model in collaboration with the IoT and edge node and maintains the privacy of data and avoids the updates of models by malicious elements. The simulation results based on the ToN-IoT dataset in the OMNeT++ environment demonstrate that FELIDS has a better detection rate with F1-Score of 0.978, Precision of 0.983 and Recall of 0.974 and low False Alarm rate of 0.021 at 50k network records. The proposed framework outperforms Suricata, Snort and Zeek by 8.06%, 9.89%, and 11.63%, 7.43%, 9.83%, and 12.99% on the F1-score, precision and recall respectively compared to the existing systems and reduces the rate of false alarm by 76.67%, 80%, and 82.5% respectively. These findings validate the claim that FELIDS is a scalable and privacy-sensitive IDS in the current IoT infrastructure. The framework can be further augmented in future work through incorporating more powerful deep federated learning models and less complex edge intelligence methods to further enhance real-time threat prediction and energy efficiency in large-scale heterogeneous IoT networks.

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